NEUROSIGHT

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Supporting evidence for Neurosight assessments

Neurosight combine two established fields of science – real-time neuroscience, and business psychology. Our innovations and claims are backed by rigorous, peer-reviewed science, and our data science is transparent.

We welcome independent data auditing and provide admin-level log-in access to our assessment platform for customers to facilitate independent verification of any claims of validity, completion time, reliability, and adverse impact.

We are the only assessment provider who offers this deep level of data access directly to customers, and we are happy to do so because we have a high degree of confidence in our methodology, and wish to instil that same level of confidence for our customers.

The application of real-time neuroscience

We define 'real-time neuroscience' as the analysis of real-time sensory data such as movement and eye-tracking to gain insight into human talent and potential – specifically as measured by behavioural and cognitive constructs including what may be termed as strengths, competencies, cognitive preferences/abilities, values, and motivations.

Our Chief Scientific Officer, Dr Craig Chapman, is a leading academic in this field. Based primarily at the University of Alberta in Canada, Dr Chapman runs the Action in Complex Environments Laboratory and facilitates our proprietary research in this field. He has additionally published a number of peer-reviewed scientific papers that have advanced understanding of how real-time sensory data can reveal how people think and feel.

Peer reviewed science has demonstrated that;

- Analysis of real-time sensory data can reveal accurate insight into how people think and feel
  - **Gavllivan, Chapman** – *Reach trajectories as a probe of decision making* (Frontiers in Neuroscience, 2014)
  - **Chapman et al** – *Separating value from selection frequency in rapid reaching biases to visual targets* (Visual Cognition, 2015)
  - **Wispinki, Gallivan, Chapman** – *Models, movements, and minds; bridging the gap between decision making and action (*Cognitive Neuroscience Review, 2018)

- Genuine and instinctive decisions have a different sensory profile to fake/forced decisions
  - **Duran et al** – *Exploring the movement dynamics of deceptions* (Frontiers in Psychology, 2013)
  - **Gavllivan, Chapman et al** – *Decision making in sensorimotor control* (Nature Reviews: Neuroscience, 2018)

- The collection and analysis of real-time sensory data is effective across a broad range of devices and input mechanisms
  - **Lavoie, Chapman et al** – *Using synchronised eye and motion tracking to determine patterns during interaction tasks* (Journal of Vision, 2018)
  - **Ouellette Zuk, Chapman** – *Evaluating the consistency of mouse-tracking derived measures of decision difficulty in reach decision tasks* (proprietary Neurosight research, 2021)

Combining the above, it is established in peer-reviewed science that the collection and analysis of real-time sensory data can reveal insight that is readily applicable in the behavioural and cognitive assessment domain. Further, the ability to differentiate between genuine and fake/forced responses overcomes a significant barrier to accurate data collection in traditional online assessment.

The application of business psychology

The founding team of Neurosight have, between them, created and validated dozens of traditional online assessments (such as situational judgement tests) for many of the world's leading organisations – including Microsoft, Barclays, Accenture, and Lloyds Banking Group. Many of these assessments are in use across the world today.

When designing and building online assessments, we anchor ourselves in best practice as we would if designing a more traditional online assessment – including job analysis, item design, validation, and implementation. We hold ourselves to the same rigorous standards of assessment design and validation as traditional online assessments.

The key difference is not how the assessment is built or validated, but that we add the real-time neuroscience data/analysis layer to gain additional insight to raise the validity of the assessment.

Our key claims and supporting evidence

Our key claims are as follows;

1) Neurosight assessments are up to 6x more accurate than traditional online assessments
2) Neurosight assessments are up to 90% faster to complete, with an average completion time of 3 minutes
3) Neurosight assessments reduce bias and promote diversity (including neurodiversity)

Claim 1: ***Neurosight assessments are up to 6x more accurate than traditional online assessments***

Situational Judgement Tests typically display construct validity of 0.1-0.2 (sometimes lower and rarely higher). Construct validity of Neurosight assessments has been evaluated across four validations (n=21, n=52, n=103, and n=109), with the first two assessing 4 constructs, and the latter two assessing 10 constructs each. Mean construct validity ranged from 0.36 to 0.65, while the top two performing items per construct (of the kind that would be deployed to a real-world assessment) averaged 0.532 – please refer to the appendix 'Construct validity' for further details.

How is this possible?

The real-time neuroscience data layer adds considerable insight into the 'how' of decision making during an online assessment. While traditional online assessments only measure the 'what' of the answer itself, Neurosight assessments are capturing and analysing hundreds of data points each second.

This allows us to draw highly accurate inferences into how people think and feel as they make decisions in an online assessment. When combined with the traditional data of the answer itself, this dramatically raises the available insight and the accuracy of the assessment.

Claim 2: ***Neurosight assessments are up to 90% faster to complete, with an average completion time of 3 minutes***

Traditional online assessments such as Situational Judgement Tests are often lengthy to complete, with each individual item typically taking up to 2 minutes to complete – with an average of 16 items,

completion times for deployed SJT's are typically in excess of 30 minutes. The average completion time for Neurosight's real-time neuroscience enhanced items is just 7 seconds.

Across an assessment measuring 8 constructs, with 16 items, the average assessment completion time for Neurosight assessments is under 3 minutes – although optional immersive content, additional guidance etc would increase that time slightly.

How is this possible?

A three minute long Neurosight assessment, enhanced with real-time sensory analysis, captures an average of 54,000 data points. In the same time period, a traditional online assessment may capture only 1 or 2 data points. This vast increase in data collection and analysis yields meaningful insight very quickly.

Neurosight assessments can be dramatically shorter in length because they attain highly valid insights at rapid pace. There is no need to make the assessments longer when high construct validity can be attained in a few minutes, thanks to the analysis of significant volumes of real-time sensory data.

Claim 3: ***Neurosight assessments reduce bias and promote diversity (including neurodiversity)***

Traditional online assessments use fixed scoring algorithms that are applied identically to each candidate. This leads to bias, since people expressing decisions outside the norm may be penalised. For example, it has been established in case law (BT v Meier, 2019) that traditional online assessments discriminate against candidates on the autistic spectrum, who may struggle to imagine hypothetical scenarios or answer with blunt honesty rather than gaming the assessment by 'playing to win'.

Across two validations (n=103 and n=109) assessing 10 constructs each, Neurosight conducted adverse impact analysis and found no meaningful difference in scores between men or women, BAME or non-BAME, and disabled or not disabled candidates. Indeed, BAME, disabled, and female scored very slightly higher than non-BAME, not disabled, and male candidates.

How is this possible?

Neurosight assessment use real-time machine learning to monitor how people make decisions in an online assessment, building an accurate picture of how decision making looks for that particular individual. We then use this data to adjust their scoring algorithms in real-time, to ensure the removal of bias. In effect, people are assessed against their own decision making profile rather than a 'norm group'.

This leads to a dramatic reduction in the kind of adverse impact you typically see in traditional online assessments. At the methodological level, this is an inherently bias-free approach to assessment.

That said, due care must still be taken with assessment design to avoid the insertion of bias at the construct level (for example, based on our existing understanding of gender-level behavioural differences, an assessment measuring competitiveness, drive, and affinity for technology would likely favour men if not balanced with other constructs such as collaboration, empathy, and conscientiousness).

Ensuring accessibility and scalability of Neurosight assessments

In partnership with the University of Alberta, Canada, we have conducted 6 validation studies (3x n=20 and 3x n=70) to ensure that real-time sensory data can be captured and analysed reliably across commonly used devices and in varied environmental settings. Our validations confirm that accurate data capture and analysis is achievable with minimal participant guidance, and across all commonly used devices.

In addition, the majority of disabilities can be accommodated without any manual intervention or adjustments, due to the nature of the individually calibrated scoring algorithms, which account for differences between individuals and provide personally tailored scoring algorithms as standard.

Frequently asked questions

How can your assessments be 90% faster and yet 6x more accurate?

We capture and score two things – what people select and, crucially, how they select it. Traditional assessments only capture the former. Across three minutes, we capture and analyse an average of 54,000 data points. Because we are capturing *how* decisions are made, we dramatically increase the available data for analysis, and can therefore reduce assessment time while still increasing accuracy.

For comparison, a traditional 16 item online assessment will capture only 16 data points (the answers) across an average of 35 minutes of assessment time.

How do you know your methodology works?

We have conducted 9 validation studies to date, involving over 500 individuals in the UK and Canada. These studies confirmed the high construct validity, short completion times, and lack of adverse impact, We balanced our validations demographically to permit adverse impact analysis to confirm our assessments are bias-free. The validity attained is far in excess of comparable validations conducted on numerous occasions by members of the founding team for traditional SJT-type online assessments.

How did you calculate your construct validity?

We approached validation with the same best-practice methods used to validate more traditional types of online assessment. Participants were asked to rate themselves on a number of gold-standard rating scales aligned to behavioural or cognitive constructs, and then completed a number of assessed items in order to determine construct validity at the item level.

How are your assessments scored?

Scoring is applied based on analysis of an individual's decision making profile, based on the real-time sensory data collected during the assessment. Through this, we build an accurate picture of how an individual responds to items they consider both easy and difficult, and use this as a benchmark to adjust the scoring algorithms. Each individual benefits from these personally tailored scoring algorithms.

What assurances can you give me that this approach to assessment really is so much better?

Unlike other assessment providers, we are happy to provide our customers with an admin-level login to our assessment platform, so they are able to independently audit any claims of validity, reliability, completion time, or adverse impact. We will always be honest and transparent with our data and wish to instil the same high level of confidence we have in our methodology for our customers.

Further to this, we welcome being trialled and compared against an existing assessment provider. We are happy to build an assessment that replicates what your existing assessment measures, deploy it, and then independently validate it in comparison to a competitor product. We are happy to do this because have an extremely high degree of confidence in our methodology.

If this approach is so much better why aren't other assessment providers doing it?

The type of data analysis we conduct isn't compatible with regular online assessment platforms, or traditional online methods of assessment such as SJT's – we built a brand new platform from the ground up to accommodate real-time neuroscience, the result of a three-year R&D programme that other assessment providers have not undertaken.

Appendix: Construct validity

Across two validations (n=103 and n=109) validity was established for 118 items across 20 common constructs. The constructs were selected based on a meta-analysis of job analysis output from 12 major employers and represent the most commonly assessed constructs across these organisations.

In addition to the individual item range and mean construct validity, the validity for the top two performing items per construct is displayed. Typically, these are the items that would be included in a final assessment and represent the 'real world' construct validity that can be expected from Neurosight assessments. The mean construct validity for the top two validated items per construct (40 items) is 0.532.

Construct	**Item-level construct validity range**	**Mean construct validity**	**Top two validated items**
Adaptive Communication	0.425 – 0.649	0.516	0.577
Collaboration	0.463 – 0.651	0.561	0.621
Conscientiousness	0.261 - 0.598	0.425	0.542
Critical Thinking	0.188 – 0.406	0.278	0.365
Customer Focus	0.264 – 0.728	0.490	0.685
Data Analysis	0.472 – 0.647	0.521	0.575
Detail Orientation	0.364 – 0.537	0.445	0.493
Disruptive Innovation	0.394 – 0.548	0.468	0.522
Drive	0.288 – 0.568	0.494	0.563
Empathic Connection	0.398 – 0.653	0.547	0.633
Flexibility	0.323 – 0.386	0.361	0.377
Judgement	0.227 – 0.564	0.406	0.558
Learning Agility	0.320 – 0.524	0.383	0.445
Numerical Mindset	0.551 – 0.757	0.656	0.729
Ownership	0.286 – 0.585	0.399	0.500
Resilience	0.266 – 0.514	0.406	0.489
Strategic Awareness	0.181 – 0.451	0.309	0.378
Tech Affinity	0.373 – 0.634	0.510	0.575
Time Optimisation	0.324 – 0.600	0.463	0.560
Verbal Mindset	0.316 – 0.573	0.449	0.537